

OSSP (Operating Systems And Systems Programming) - 25CS2104E

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1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[50];

    printf("Enter Linux Command: ");
    scanf("%s", cmd);

    pid_t pid = fork();

    if (pid < 0)
    {
        printf("Fork Failed\n");
        return 1;
    }
    else if (pid == 0)
```

```

{
    printf("\nChild Process\n");
    printf("Child PID : %d\n", getpid());
    printf("Parent PID: %d\n", getppid());

    execlp(cmd, cmd, NULL);

    perror("Command Execution Failed");
    exit(1);
}
else
{
    printf("\nParent Process\n");
    printf("Parent PID : %d\n", getpid());
    printf("Child PID : %d\n", pid);

    wait(NULL);

    printf("\nChild Process Completed\n");
}

return 0;
}

```

Output:

```

gssgvy@Gyanadeep:~$ cd ~/OSLab
gssgvy@Gyanadeep:~/OSLab$ gedit practical1.c &
[1] 522
gssgvy@Gyanadeep:~/OSLab$
** (gedit:522): WARNING **: 13:17:11.985: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:522): WARNING **: 13:17:11.986: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

[1]+  Done                  gedit practical1.c
gssgvy@Gyanadeep:~/OSLab$ mv practical1.c Practical-01/
gssgvy@Gyanadeep:~/OSLab$ cd Practical-01
gssgvy@Gyanadeep:~/OSLab/Practical-01$ gedit practical1.c &
[1] 873
gssgvy@Gyanadeep:~/OSLab/Practical-01$
** (gedit:873): WARNING **: 13:17:53.157: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:873): WARNING **: 13:17:53.157: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

[1]+  Done                  gedit practical1.c
gssgvy@Gyanadeep:~/OSLab/Practical-01$ cd ~/OSLab/Practical-01
gssgvy@Gyanadeep:~/OSLab/Practical-01$ ./practical1
-bash: ./practical1: No such file or directory
gssgvy@Gyanadeep:~/OSLab/Practical-01$ gcc practical1.c -o practical1
gssgvy@Gyanadeep:~/OSLab/Practical-01$ ./practical1
Enter Linux Command: ls

Parent Process
Parent PID : 1192
Child PID : 1193

Child Process
Child PID : 1193
Parent PID: 1192
practical1 practical1.c

Child Process Completed
gssgvy@Gyanadeep:~/OSLab/Practical-01$ |

```